

PRANAV RAMESH

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[LinkedIn](#) | [GitHub](#)

Education

University of Waterloo Bachelor of Applied Science in Computer Engineering, Co-operative Program	<i>Waterloo, ON</i> Expected 2031
VIBGYOR High School, Horamavu CBSE Class XII: 96.2%	<i>Bengaluru, India</i>

Technical Skills

Programming	C/C++, Python, JavaScript, React, React Native, MySQL
Embedded Systems	ESP32, Raspberry Pi, Embedded C/C++, Interfacing, IoT connectivity, Firebase
Electronics	Digital/Power Electronics, PCB Design, high-frequency switching, gate-drive design
Development	Linux, Git/GitHub, Vim, circuit simulation, hardware debugging
Hardware	PCB prototyping, soldering, oscilloscopes, multimeters, circuit debugging

Projects

Phase Locked Loop Solid-State Tesla Coil [\[GitHub\]](#) 2022 – Present

- Designed a 1–2 kW PLL SSTC targeting 250 kHz with a 325 VDC full-bridge power stage.
- Designed multi-layer KiCad PCBs and schematics for digital logic, gate drive and power switching.
- Programmed an Attiny85 in C++ to process MIDI files and generate amplitude-modulated output. [\[Demo\]](#)

FarmerHelper – Remote IoT Irrigation System [\[GitHub\]](#) 2020 – 2024

- Designed and deployed an IoT irrigation system running continuously on an agricultural installation.
- Programmed a cellular ESP32 to control valves and motor pump while synchronizing with Firebase.
- Built a React Native app with Firebase Authentication, Realtime Database, and OTP authentication.
- Implemented regional-language support using an extensible multi-language application architecture.

Smart Cars – Featured in The Times of India [\[Article\]](#) 2018

- Featured in The Times of India’s “Young Student Innovators” and invited by Freudenberg to present the project.
- Built Wi-Fi-controlled smart cars using Raspberry Pi with live video streaming.
- Integrated sensing, control, and video streaming into a mobile robotics platform.

AI Home Intrusion Detection System [\[GitHub\]](#) 2019

- Built a Raspberry Pi security system combining IR motion detection with AI facial recognition to flag unauthorized access.

Automated Home Water Management System 2016 – 2018

- Designed a water level controller using embedded C++ to prevent water wastage.

Activities & Achievements

- Inline speed skating: Won 2 District Championship gold medals, qualified nationally, and placed 11th at a semi-national competition.
- Designed and taught IoT/electronics workshops for university and school students and at an orphanage.
- Model United Nations delegate – developed public speaking, negotiation, and leadership skills through committee sessions.
- Long distance cycling: Cycled 200km in 8 hours after extensive training and persevering.